Nikolai Slavov

Contact Information

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Education

Princeton University, Princeton, USA

Ph.D. in Molecular and Quantitative Cell Biology — **06.2010**

M.A. in Molecular Biology — **02.2006**

Dissertation: Universality, specificity and regulation of *S. cerevisiae* growth rate response in different carbon sources and nutrient limitations

Mentor: David Botstein

Massachusetts Institute of Technology (MIT), Cambridge, USA

B.S. in Biology — **06.2004**

Academic Appointments

Northeastern University, Boston, USA

Department of Bioengineering

Assistant Professor — **2015–present**

Broad Institute of Harvard and MIT, Cambridge, USA

Proteomics Platform — **2015–present**

Harvard University, Cambridge, USA

Departments of Systems Biology and Statistics — **2014–2015**

Massachusetts Institute of Technology (MIT), Cambridge, USA

Departments of Biology and Physics

Postdoc in the van Oudenaarden Laboratory — **2011–2013**

Awards & Fellowships

✓ NIH Director's New Innovator Award ($\$ \; 1,500,000 + 852,750$) — **2016**

✓ NEU Tier I Award ($\$ \; 50,000$) — **2016**

✓ Broad Institute of Harvard and MIT SPARC Award ($\$ \; 100,000$) — **2014**

✓ Princeton University Dean's Award — **2010**

✓ IRCSET Postgraduate Research Fellowship ($72,000$ €) — **2007**

✓ Finalist in the Young European Entrepreneur Competition — **2006**

✓ Princeton Graduate Fellowship ($\$ \; 45,000$) — **2004**

✓ MIT Undergraduate Fellowship ($\$ \; 125,000$) — **2001**

✓ Eureka Fellowship for Academic Excellence ($10,000$ €) — **2000**

✓ Bronze Medal in the 31^{st} International Chemistry Olympiad (*IChO*) — **1999**

✓ National Diploma for Exceptional Achievements in Chemistry — **1999**

Professional Activities

- **Peer reviewer:** PNAS, Molecular Biology of the Cell, PLoS, Nucleic Acids Research, Bioinformatics, Annals of Applied Statistics, Cell Reports, eLife

Selected Peer Reviewed Publications

✉ – Corresponding author; **[OA]** – Open Access article

2016

- ✓ Klionsky, *et al.*, **Slavov N.**, *et al.*
  Guidelines for the use and interpretation of assays for monitoring autophagy
  Autophagy, vol. 12, issue 1, 1–222

2015

- ✓ **Slavov N.**✉
  Making the most of peer review
  eLife, 4:e12708 **[OA]**
- ✓ **Slavov N.**✉, Semrau S., Airoldi E.M., Budnik B., van Oudenaarden A.
  Differential stoichiometry among core ribosomal proteins
  Cell Reports, vol. 13, issue 5, 865–873 **[OA]**
- ✓ Alvarez JR, Zhiqiang B, Xu D, Yuan B, Lo KA, Yoon MJ, Lim YC, Knoll M, **Slavov N.**, *et al.*
  De-Novo Reconstruction of Adipose Tissue Transcriptomes Reveals Long Non-coding RNA Regulators of Brown Adipocyte Development
  Cell Metabolism, vol. 21, issue 5, 764–776

2014

- ✓ Malioutov D.✉, **Slavov N.**✉
  Convex Total Least Squares
  Journal of Machine Learning Research, W&CP, vol. 32, issue 1, 109–117 **[OA]**
- ✓ **Slavov N.**✉, Budnik B., Schwab D., Airoldi E., van Oudenaarden A.✉
  Constant Growth Rate Can Be Supported by Decreasing Energy Flux and Increasing Aerobic Glycolysis
  Cell Reports vol. 7, issue 3, 705–714 **[OA]**

2013

- ✓ **Slavov N.**✉, Carey, J., Linse, S.✉
  Calmodulin transduces Ca^{+2} oscillations into differential regulation of its target proteins
  ACS Chemical Neuroscience, vol. 4, 601–612 **[OA]**
- ✓ **Slavov N.**✉, Botstein D.✉
  Decoupling Nutrient Signaling from Growth Rate Causes Aerobic Glycolysis and Deregulation of Cell Size and Gene Expression
  Molecular Biology of the Cell, vol. 24, issue 2, 157–168 **[OA]**

2012

- ✓ **Slavov N.**✉, van Oudenaarden A.✉
  How to Regulate a Gene: To Repress or to Activate?
  Molecular Cell, vol. 46, issue 5, 551–552 **[OA]**

✓ **Slavov N.**✉, Airoldi E.M., van Oudenaarden A., Botstein D.✉
A Conserved Cell Growth Cycle Can Account for the Environmental Stress Responses of Divergent Eukaryotes
Molecular Biology of the Cell, vol. 23, no. 10, 1986–1997 **[OA]**

2011

2011

✓ **Slavov N.**✉, Macinskas J., Caudy A., Botstein D.✉
Metabolic Cycling without Cell Division Cycling in Respiring Yeast
PNAS, vol. 108, no. 47, 19090–19095 **[OA]**

✓ **Slavov N.**✉, Botstein D.✉
Coupling among Growth Rate Response, Metabolic Cycle and Cell Division Cycle in Yeast
Molecular Biology of the Cell, vol. 22, no. 12, 1997–2009 **[OA]**

2010

2010

✓ **Slavov N.**✉
Inference of Sparse networks with Unobserved Variables. Application to Gene Regulatory Networks
Journal of Machine Learning Research, W&CP, vol. 9, 757–764 **[OA]**

✓ **Slavov N.**✉, Botstein, D.
Universality, Specificity and Regulation of *S. cerevisiae* Growth Rate Response in Different Carbon Sources and Nutrient Limitations
Dissertation, Princeton University **[OA]**

✓ Silverman SJ., **Slavov N.**, Petti A., Parsons L., Briehof R., Thiberge S., Zenklusen D., Gandhi SJ., Larson D., Singer R., Botstein D.✉
Metabolic cycling in single yeast cells from unsynchronized steady-state populations limited on glucose or phosphate
PNAS, vol. 107, no. 15, 6946–6951 **[OA]**

2009 and before

2009 and before

✓ **Slavov N.**✉, Dawson KA. (2009)
Correlation Signature of the Macroscopic States of the Gene Regulatory Network in Cancer
PNAS, vol. 106, no. 11, 4079-4084 (“In This Issue” article) **[OA]**

✓ Tagkopoulos I.✉, **Slavov N.**✉, Kung S.✉ (2005)
Multi-Class Biclustering and Classification Based on Modeling of Gene Regulatory Networks
5^{th} IEEE Symposium on Bioinformatics and Bioengineering (BIBE’05).

PUBLISHED PREPRINTS

▷ **Slavov N.**✉, Franks A., Airoldi E.M. (2015) Extensive post-transcriptional regulation across human tissues, *bioRxiv*, DOI: http://dx.doi.org/10.1101/020206 **[OA]**

▷ **Slavov N.**✉, Botstein, D., Caudy A. (2014) Extensive regulation of metabolism and growth during the cell division cycle, *Under review*, arXiv:1406.3398, DOI: 10.1101/005629 **[OA]**

Popular Science Publications

- ▷ **Slavov N.**✉ (2015) Making the most of peer review
  eLife, Link
- ▷ **Slavov N.**✉ (2014) Accomplishments Over Accolades!
  The Scientist, Link
- ▷ **Slavov N.**✉ (2014) Discoveries lie hidden behind the façade of popular assumptions
  Cell Reporter, Link
- ▷ **Slavov N.**✉ (2013) The Best Projects Are Least Obvious
  The Scientist, Link
- ▷ **Slavov N.**✉ (2013) A Wonderful Christmas with Mysterious Oscillations
  PubChase, Link
- ▷ **Slavov N.**✉ (2012) The Mission of MIT
  The MIT Tech, Link

Research Talks

- Principles of ribosome-mediated translational regulation
  Systems Biology Seminar, Boston University **Oct.13.2016**
- Differential stoichiometry among core ribosomal proteins
  CSHL Translational Control, Cold Spring Harbor Laboratory **Sept.08.2016**
- Ribosome-mediated translational regulation: Direct methods for quantifying the ribosome code
  Center for Systems Biology, Columbia University **Apr.20.2016**
- Coordinating metabolism and ribosome-mediated translational regulation
  Department of Biology, New York University **Apr.19.2016**
- Principles of ribosome-mediated translational regulation
  Center for Interdisciplinary Research on Complex Systems **Apr.12.2016**
- Quantifying homologous proteins and proteoforms
  US HUPO Conference **Mar.14.2016**
- Coordination among cell growth, cell division and gene expression
  Broad Institute of MIT and Harvard, YSB **Dec.9.2015**
- Coordinating cell growth and metabolism with ribosome-mediated translational regulation
  Northeastern University, Biology Department Colloquium **Oct.19.2015**
- Quantifying protein isoforms
  Broad Institute of MIT and Harvard **Sep.14.2015**
- Variable stoichiometry among core ribosomal proteins
  2014 ASCB/IFCB Annual Meeting **Dec.8.2014**
- Direct quantification of highly homologous proteins by mass–spectrometry indicates physiological changes

in the protein composition of the ribosome
Broad Institute of MIT and Harvard, CBBO **Jun.25.2014**

- Variable stoichiometry among core ribosomal proteins
  Broad Institute of MIT and Harvard **Jun.19.2014**

- Quantification of translational regulation of homologous proteins with unprecedented accuracy
  Harvard HMS Systems Biology Theory Lunch **Mar.21.2014**

- Exponentially growing cells can support a constant growth rate by very different metabolic strategies
  Fourth Annual Physical Sciences in Oncology Meeting, NCI **Apr.18.2013**

- Experimentally identified trade-offs of aerobic glycolysis
  Center of Cancer Systems Biology at Tufts **Dec.18.2012**

- Trade-offs of aerobic glycolysis
  Harvard HMS Systems Biology Theory Lunch **Sep.12.2012**

- The dynamics of exponential growth
  GSA Yeast Meeting, Princeton University **Aug.5.2012**

- The dynamics of exponential cell growth at constant growth rate
  MIT Physical Sciences-Oncology Center **Jun.11.2012**

- The dynamics of exponential growth
  Broad Institute of Harvard and MIT **Apr.17.2012**

- Metabolic cycling without cell division cycling
  Bauer Forum at Harvard FAS Systems Biology **Apr.15.2011**

- Inference of Sparse Networks with Unobserved Variables. Application to Gene Regulatory Networks
  International Conference on Artificial Intelligence and Statistics **May.14.2010**

- Dynamics in the calcium-calmodulin signaling network
  International Conference on Chaos and Nonlinear Dynamics **Jan.8.2006**

TEACHING & RESIDENTIAL ADVISING

Mathematical Methods for Engineers, Northeastern University **2016 – 2017**
Introduction to major concepts and methods in linear algebra differential equations

Method and Logic in Quantitative Biology, Princeton University **2009 – 2010**
Introduction to major ideas and concepts in quantitative biology based on primary literature

Residential Graduate Student at Forbes College, Princeton University **2009 – 2010**
Advised students; organized language tables, sport events and trips

Integrated Quantitative Introduction to Natural Sciences, Princeton University **2006 – 2010**
Interdisciplinary curriculum including physics, computer science, biochemistry, genetics, physiology and emphasizing quantitative problem solving and the connections among these disciplines

Terrascope, MIT **2002 – 2003**

Selected for two subsequent years to lead undergraduate student groups working on complex real-world problems

Biochemistry, MIT **2003 – 2004**

Taught precepts and helped undergraduate students prepare for examinations